

DAVID LÓPEZ DIEGO

Materials Scientist | Nanomaterials Characterization & Scientific Data Analysis

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PROFESSIONAL SUMMARY

Physicist and PhD candidate with over four years of experience in advanced materials characterization, nanoscale metrology, scientific instrumentation and experimental data analysis. My work spans nanomaterials, semiconductors, biosensors and thermoelectric materials, combining laboratory measurements with computational workflows for data processing and technical reporting. I develop reproducible characterization procedures, support instrument calibration and quality documentation, and translate experimental results into traceable datasets and scientific outputs. I have co-authored of 10+ peer-reviewed publications and contributed to multidisciplinary and industry-linked R&D projects.

TECHNICAL SKILLS

Materials Characterization: AFM, SEM-EDX, TEM, XRD, Raman, FTIR, XPS, FIB lithography, optical & fluorescence microscopy, HPLC-MS, ICP

Data Analysis & Programming: Python, Scikit-learn, TensorFlow, R, Excel, Origin, ImageJ, Git, Mathematica

Instrumentation & Lab Ops: Calibration, preventive maintenance, data acquisition, cleanroom environment, bacterial cultures, equipment logs

Quality & Documentation: GLP/GMP principles, ISO 9001, workflow standardization, procurement and budget tracking

Additional Tools: SQL, CST Studio Suite, AutoCAD, C++, NI LabVIEW, 3D printing, Power BI

PROFESSIONAL EXPERIENCE

Materials Scientist - Institute of Micro and Nanotechnology, IMN-CNM, CSIC | Madrid, Spain | Feb 2022 - Present

- Develop AFM-based methods for nanoscale characterization of nanomaterials, semiconductors, biosensors and biological systems. Establish standardized measurement workflows adopted by 4 research groups, ensuring consistency across multidisciplinary projects.
- Operate, calibrate and maintain 10+ characterization techniques (AFM, SEM-EDS, TEM, FIB, XRD, HPLC-MS, Raman, XPS, ICP, FTIR). Provide instrumental support to 5 research teams and external collaborators, ensuring consistent data quality for ongoing R&D projects.
- Design end-to-end experimental workflows, defining measurement protocols, data structures and reporting criteria to ensure traceability from sample preparation to final results.
- Co-author 10+ peer-reviewed publications in materials characterization, nanoscience, biosensors and applied research.
- Contribute to national and industry-linked R&D initiatives, including PERTE Chip (semiconductor characterization), qNanoFoPho (photonic nano-structures), and biosensor development for early autism detection, providing experimental measurements.
- Perform electrical and thermal transport characterization of thermoelectric materials using Hall, LFA, DSC, TGA, Seebeck, NanoTR and PicoTR measurement systems.
- Maintain ISO 9001-certified laboratory operations through GLP/GMP principles: manage equipment logs, calibration records, SOP documentation and audit preparation, ensuring successful annual quality system reviews.
- Develop Python tools for scientific data analysis, numerical modelling and technical reporting, replacing repetitive manual processing with standardized and reproducible procedures.
- Manage project-related procurement, supplier coordination and budget tracking to support timely delivery of research milestones.
- Train 5+ junior researchers on analytical techniques, sample preparation and instrumentation workflows, enabling them to conduct independent measurements and data analysis within research projects.

Research Assistant, Materials Physics - SMAP Group, Surfaces and Porous Materials | Valladolid, Spain | Jan 2021 - Sep 2021

- Synthesized and characterized 10+ mixed-matrix membrane and polymer formulations for gas-separation applications, optimizing permeability/selectivity/performance trade-offs for CO₂ and greenhouse-gas capture use cases.
- Managed laboratory operations, instrument maintenance and safety protocols in a materials research environment.

EDUCATION

PhD Candidate in Condensed Matter Physics, Nanoscience and Biophysics - Autonomous University of Madrid, Spain | Expected: 2027

MSc in Physics (Materials Physics) - University of Valladolid | 2021

BSc in Physics - University of Valladolid | 2020

CERTIFICATIONS & TRAINING

- Data Science, Machine Learning & GenAI** (Python, R, SQL) – Harvard (2024) | Columbia (2025) | Udemy (2025)
- Computational Physics** (Mathematica, C++, NI LabVIEW) – University of Valladolid (2021-2022)

LANGUAGES

Spanish: Native | English: C1